

GeoVault AI

Dudhichua OCP

Mine Annual Performance & Operational Report

Financial Year 2020-21

Report ID	GV-DUDHICHUA-OCP-202021-ANNUAL
Mine ID	GV005
Subsidiary	NCL
State	Uttar Pradesh
Coalfield	Singrauli Coalfields
Mine Type	Opencast
Dataset Classification	SYNTHETIC_DEMO
Prepared For	GeoVault AI RAG / analytics demonstration

Important: This is a synthetic demonstration report generated from the GeoVault synthetic dataset. It is designed for ingestion, retrieval, analytics, evidence citation and report-generation testing. It is not an official Coal India Limited, CMPDI or subsidiary publication.

1. Executive Summary

During FY 2020-21, Dudhichua OCP recorded actual coal production of 18.20 MT against a target of 21.00 MT, corresponding to achievement of 86.67%. The production variance was -2.80 MT. Year-on-year production growth was -4.71% compared with the preceding financial year. The corresponding offtake was 17.67 MT against an offtake target of 20.68 MT, with offtake achievement of 85.44%. This production/offtake profile provides the primary operational baseline for GeoVault analysis.

The principal recorded operational constraint for FY 2020-21 was Monsoon rainfall reduced effective working days and equipment utilization.. The dataset records 9 rainfall-impact days and 152 hours of equipment or operational downtime. Equipment availability was 93.6%, while available manpower was 915. These indicators should be interpreted together rather than treating any single metric as a definitive causal explanation. GeoVault should surface the stated cause alongside the supporting operational indicators so that an analyst can inspect the evidence.

2. Mine Profile

Attribute	Value
Mine	Dudhichua OCP
Mine ID	GV005
Subsidiary	NCL
State	Uttar Pradesh
Coalfield	Singrauli Coalfields
Mine Type	Opencast

Dudhichua OCP is represented in GeoVault as a structured mine entity linked to annual performance records. The profile establishes stable metadata—subsidiary, state, coalfield and mine type—that can be used for filtering, authorization scope and retrieval. Annual metrics are stored separately so that queries such as production trend, target achievement or year-over-year growth can be computed across financial years.

3. Production Performance

Metric	Value
Production Target (MT)	21.00
Actual Production (MT)	18.20
Production Variance (MT)	-2.80
Achievement	86.67%
Previous-Year Production (MT)	19.10
YoY Growth	-4.71%
Average Daily Production ('000 t/day)	49.86

During FY 2020-21, Dudhichua OCP recorded actual coal production of 18.20 MT against a target of 21.00 MT, giving achievement of 86.67%. Year-on-year production growth was -4.71%. The variance is retained as an explicit metric for analytical ranking and investigation.

4. Target vs Achievement

Measure	Target	Actual	Achievement
Coal Production (MT)	21.00	18.20	86.67%
Coal Offtake (MT)	20.68	17.67	85.44%

Measure	Target	Actual	Achievement
OB Removal (McuM)	45.36	40.04	See annual record

For FY 2020-21, coal production achievement was 86.67%, while offtake achievement was 85.44%. Production variance was -2.80 MT. GeoVault should preserve these measures independently so that an analyst can distinguish production performance from dispatch performance.

5. Offtake & Dispatch

Actual offtake for Dudhichua OCP was 17.67 MT against an offtake target of 20.68 MT, corresponding to 85.44% achievement. Offtake is stored separately from production because coal produced at the mine and coal dispatched/off-taken are operationally distinct measures. This distinction supports precise SQL and RAG answers.

6. Overburden & Stripping Ratio

Indicator	Target	Actual
Overburden Removal (McuM)	45.36	40.04
Stripping Ratio	2.16	2.20
Rainfall Impact Days	—	9

The FY 2020-21 record shows an actual stripping ratio of 2.20 against a target of 2.16, with 9 rainfall-impact days. Actual overburden removal was 40.04 McuM against 45.36 McuM. These indicators provide operational context but should not be treated as automatic proof of causality.

7. Equipment & Productivity

Indicator	Value
HEMM Availability	93.50%
Equipment Availability	93.60%
Downtime Hours	152
Manpower Available	915
Average Daily Production ('000 t/day)	49.86
Equipment Constraint	N/A

Equipment availability was 93.60% and HEMM availability was 93.50%. The record contains 152 downtime hours and manpower availability of 915. These measures are retained as supporting evidence for production-performance analysis.

8. Operational Constraints

The primary operational cause/constraint recorded for FY 2020-21 is **Monsoon rainfall reduced effective working days and equipment utilization..** Supporting indicators include 9 rainfall-impact days and 152 downtime hours. GeoVault should distinguish this recorded cause from analytical inference and cite the underlying mine-year evidence.

9. Geological Observations

Monsoon drainage and sump management.

The geological observation is stored as a narrative field for semantic retrieval. In a production deployment, such observations should be linked to the originating geological report, section/page and relevant mine-plan or exploration evidence.

10. Safety Performance

Indicator	Value
Safety Incident Count	9
LTIFR	0.42
Fatalities	0

Safety performance for FY 2020-21 includes 9 recorded safety incidents, LTIFR of 0.42, and 0 fatalities in the synthetic record. These indicators are presented separately from production and environmental performance.

11. Environment & Mine Reclamation

Indicator	Value
Reclamation Area (ha)	18.00
Mine Water Treated (ML)	8.55
Dust Compliance	94.00%

Environmental indicators include 18.00 ha of reclamation, 8.55 ML of mine water treated and 94.00% dust compliance. These measures can support authorized environmental dashboards and cross-year trend analysis.

12. Key Operational Event

No separately recorded event was included for this mine-year.

13. Management Observations & Recommended Follow-up

Recommended analytical follow-up is to compare production achievement (86.67%) and offtake achievement (85.44%) with equipment availability (93.60%), downtime (152 hours), stripping ratio (2.20), and the stated operational constraint. Any management recommendation produced by GeoVault should remain a draft for human review and retain evidence references.

14. Source / Provenance Statement

All numeric and narrative values in this demonstration report originate from the GeoVault synthetic demonstration dataset. The corpus is intended for testing ingestion, chunking, SQL analytics, RAG retrieval, evidence grounding, citation, topic identification and report generation. It must not be represented as official CIL/CMPDI data. The underlying records are marked SYNTHETIC_DEMO.